(Please write your Enrolment No. immediately) Enrolment No. _____

MID TERM EXAMINATION

B.TECH PROGRAMMES, 4TH Semester, March, 2025

Paper Code: CIC-212 **Subject: Programming in Java**

Time: 1½ Hrs. **Max. Marks: 30**

Note: Attempt Q. No. 1 which is compulsory and any two more questions from remaining.

Q. No.	Question (Attempt any 5 parts of Q1)	Max. Marks	CO(s)
1 (a)	What is wrapper class and name the wrapper class methods to convert string objects to primitive int.	2	CO 2
1 (b)	What is the sandbox model in java?	2	CO 3
1 (c)	What are the key characteristics of Java that make it platform-independent?	2	CO 1
1 (d)	What is the role of static keyword in java language?	2	CO 3
1 (e)	Difference between super and this keyword.	2	CO 2
1 (f)	What is garbage collection in Java, and how does it work?	2	CO 1
1 (g)	What is the difference between abstract class and interface?	2	CO 4
2 (a)	What is JVM and explain various components of JVM architecture? OR Explain the uses of the final keyword with suitable Java program.	5	CO 1
2 (b)	Explain the types of constructors in Java with suitable example. OR What is exception handling in Java and its advantages? Differentiate between throw and throws keyword in Java?	5	CO 2
3 (a)	Explain the life cycle of an Applet with suitable example. OR What are control flow statements in java, explain with suitable examples?	5	CO 3
3 (b)	What is an interface in java and how it is different from a class? OR Differentiate between method overloading & method overriding with an example.	5	CO 4